Sant Dnyaneshwar Shikshan Sanstha's

Annasaheb Dange College of Engineering and Technology (ADCET), Ashta

Department of CSE (IOT and CSBT)

Quality Laboratory Manual

Introduction to Cyber Security Lab

2ICPC205

Second Year

Sem-IV

AY 2026-27

Course Instructor
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Experiment No. 3

Title of Experiment: Capture and Analyze Network Packets and Identify Network Protocols Using Cisco Packet Tracer.

Aim of Experiment: To capture and analyze network packets using Cisco Packet Tracer Simulation Mode and identify the protocols involved in network communication.

Course Outcomes (CO)

CO1: Understand the basic concepts of computer networks and network devices.
CO2: Design and simulate a simple LAN using Cisco Packet Tracer.
CO3: Verify network connectivity using simulation tools.

System Requirements:
-Cisco Packet Tracer (Version 8.2 or above)
-Computer with Windows/Linux Operating System

Theory:-

Whenever devices communicate over a network, data is encapsulated into packets and transmitted through different layers of the TCP/IP protocol stack. Each protocol performs a specific function during communication.

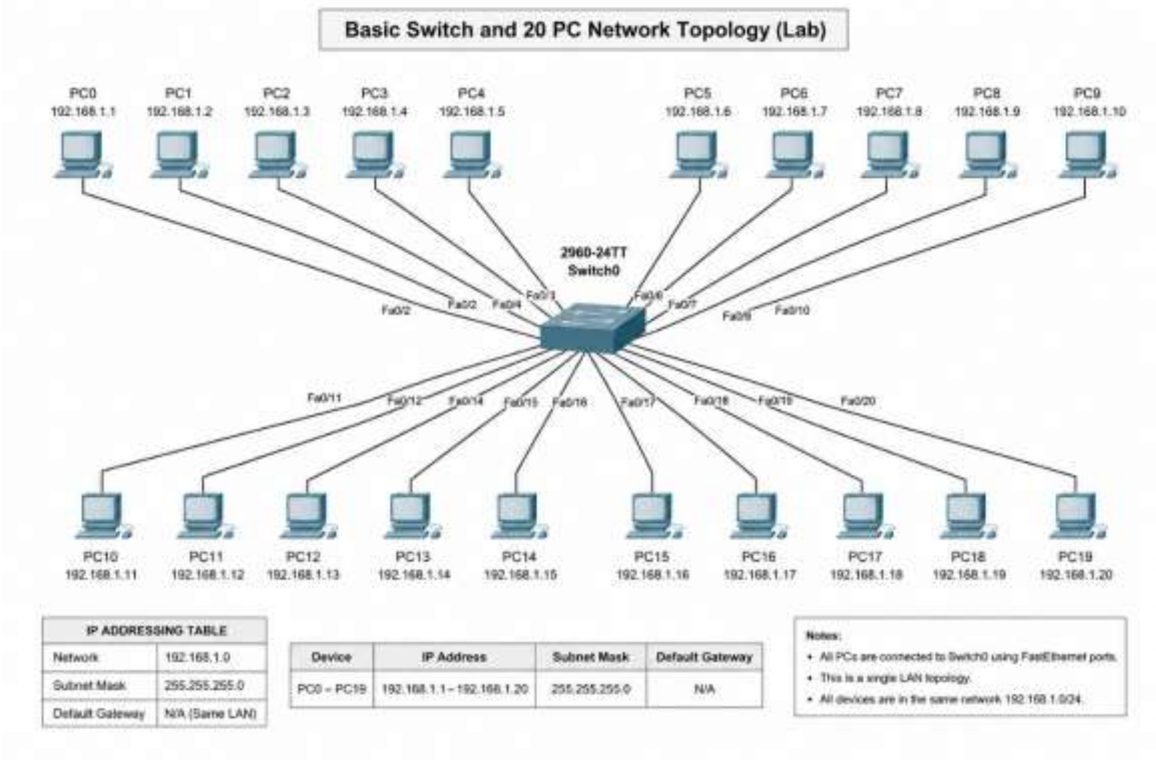
Cisco Packet Tracer provides a **Simulation Mode**, allowing users to observe packet transmission step by step. Students can inspect protocol details such as ARP requests, ICMP echo messages, Ethernet frames, and IP packets.

This experiment helps students understand how different protocols interact during communication.

Protocol	Purpose
Ethernet	Data transmission within a LAN
ARP	Resolves IP address to MAC address
IPv4	Logical addressing and packet routing
ICMP	Used by the Ping command for connectivity testing

Network Topology

- **1 Cisco 2960-24TT Switch**
- **20 PCs**
- **Copper Straight-Through cables**
- **Network: 192.168.1.0/24**



IP Addressing Table

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
PC5	192.168.1.6	255.255.255.0
PC6	192.168.1.7	255.255.255.0
PC7	192.168.1.8	255.255.255.0
PC8	192.168.1.9	255.255.255.0
PC9	192.168.1.10	255.255.255.0
PC10	192.168.1.11	255.255.255.0
PC11	192.168.1.12	255.255.255.0
PC12	192.168.1.13	255.255.255.0

Device	**IP Address**	**Subnet Mask**
PC13	**192.168.1.14**	**255.255.255.0**
PC14	**192.168.1.15**	**255.255.255.0**
PC15	**192.168.1.16**	**255.255.255.0**
PC16	**192.168.1.17**	**255.255.255.0**
PC17	**192.168.1.18**	**255.255.255.0**
PC18	**192.168.1.19**	**255.255.255.0**
PC19	**192.168.1.20**	**255.255.255.0**

Default Gateway: Not Required (Single LAN)

Procedure

Procedure

Step 1: Open Existing Topology

Open the Packet Tracer file created in Experiment 2.

Step 2: Verify IP Configuration

Device	**IP Address**
PC0	192.168.1.1
PC19	192.168.1.20

Step 3: Switch to Simulation Mode

1. Click Simulation (bottom-right corner).
2. Clear any existing events.
3. Keep only the following protocols enabled:
    - ARP
    - ICMP
    - Ethernet
    - IPv4

Step 4: Generate Network Traffic

Open **PC0 → Desktop → Command Prompt**

Type:

```
ping 192.168.1.20
```

Step 5: Observe Packet Flow
Click Capture/Forward repeatedly.
Observe:

- ARP Request
- ARP Reply
- ICMP Echo Request
- ICMP Echo Reply

Step 6: Inspect Packet Details
Click each packet and observe:
OSI Model Tab

- Layer 2
- Layer 3
- Layer 4

Inbound PDU Details
Observe:

- Source MAC Address
- Destination MAC Address
- Source IP Address
- Destination IP Address
- Protocol Type

Step 7: Repeat
Perform packet capture between different PCs.
Example:

- PC5 → PC10
- PC8 → PC17
- PC2 → PC14

Observe whether ARP packets appear again.

Observations

Communication	Protocols Observed	Result
PC0 → PC19	ARP, ICMP, Ethernet, IPv4	Successful
PC5 → PC10	ARP, ICMP	Successful
PC8 → PC17	ICMP	Successful
PC2 → PC14	ICMP	Successful

Packet Analysis Table

Parameter	**Observation**
Source IP	__________
Destination IP	__________
Source MAC	__________
Destination MAC	__________
Protocol	__________
Packet Type	Request / Reply

Result

Network packets were successfully captured and analyzed using Cisco Packet Tracer Simulation Mode. Ethernet, ARP, IPv4, and ICMP protocols were identified, and their roles in network communication were observed.

Learning Outcomes

After completing this experiment, students will be able to:

- Understand packet flow in a LAN.
- Identify Ethernet, ARP, IPv4, and ICMP protocols.
- Analyze packet encapsulation using the OSI model.
- Interpret packet details in Cisco Packet Tracer.
- Verify communication using Simulation Mode.

Conclusion :-

Expected Oral Questions –

1 What is a network packet? CO1
2 What is packet encapsulation? CO1
3 What is the purpose of ARP? CO1
4 What is the function of ICMP? CO1
5 What is the role of the IPv4 protocol? CO1
6 Which protocol carries data within a LAN? CO1
7 Why does ARP occur before the first ping? CO3
8 Why is ARP not required for every subsequent ping? CO3
9 What information is available in the PDU Details window? CO2

Interview Questions:-

10	What is the difference between an ARP Request and an ARP Reply?	CO3
11	Which protocol is used by the ping command?	CO1
12	What is the purpose of Simulation Mode in Packet Tracer?	CO2
13	How can you identify the source and destination IP addresses in a packet?	CO2
14	What happens if the destination IP address is incorrect?	CO3